LLMs and Tools

Deep Learning

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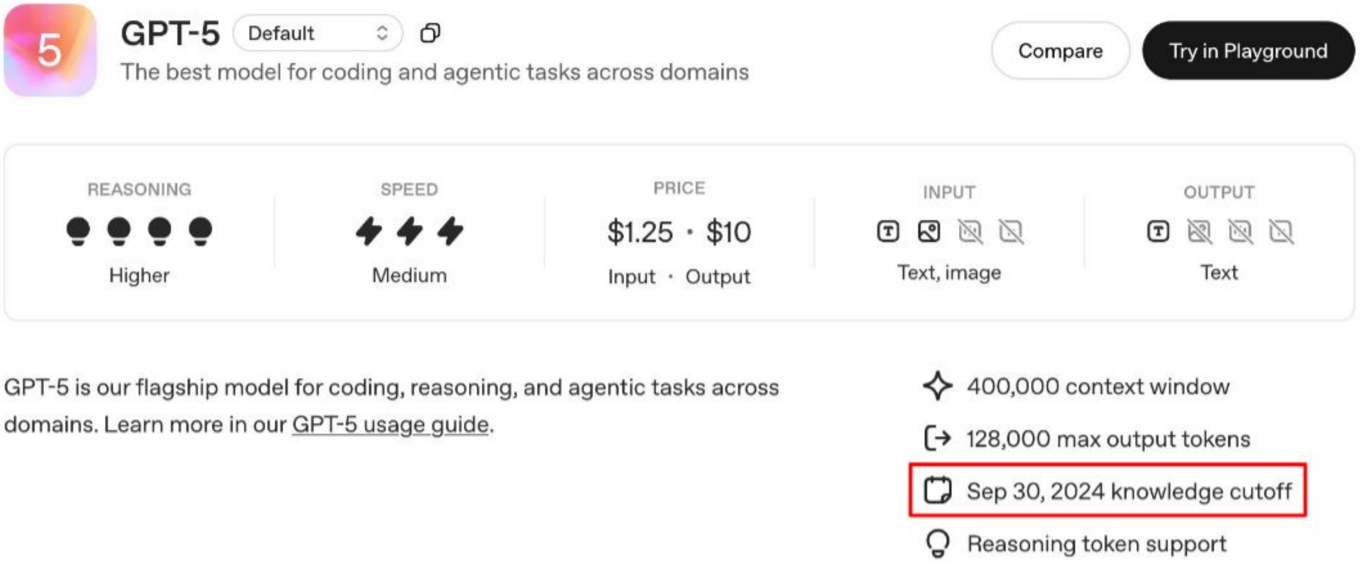
Lambda, HSE

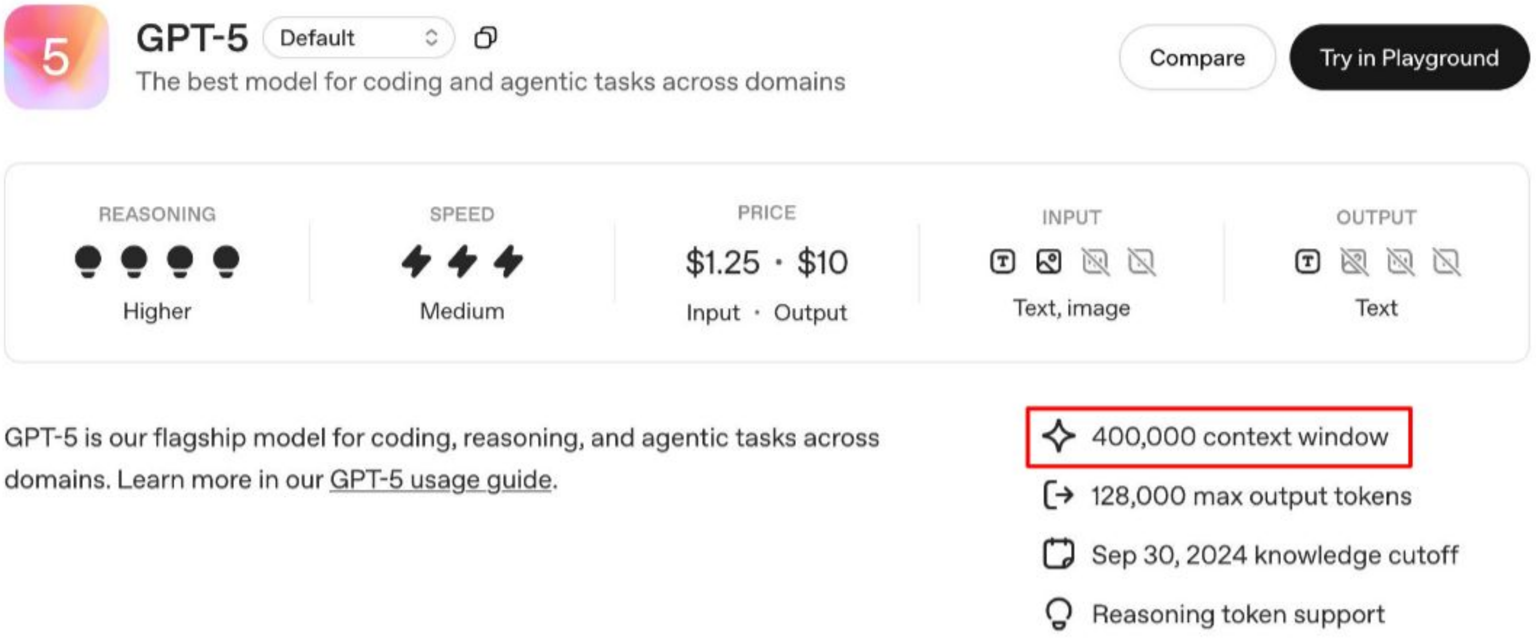
Motivation

- **Knowledge** of LLM **constrained** to pretraining data



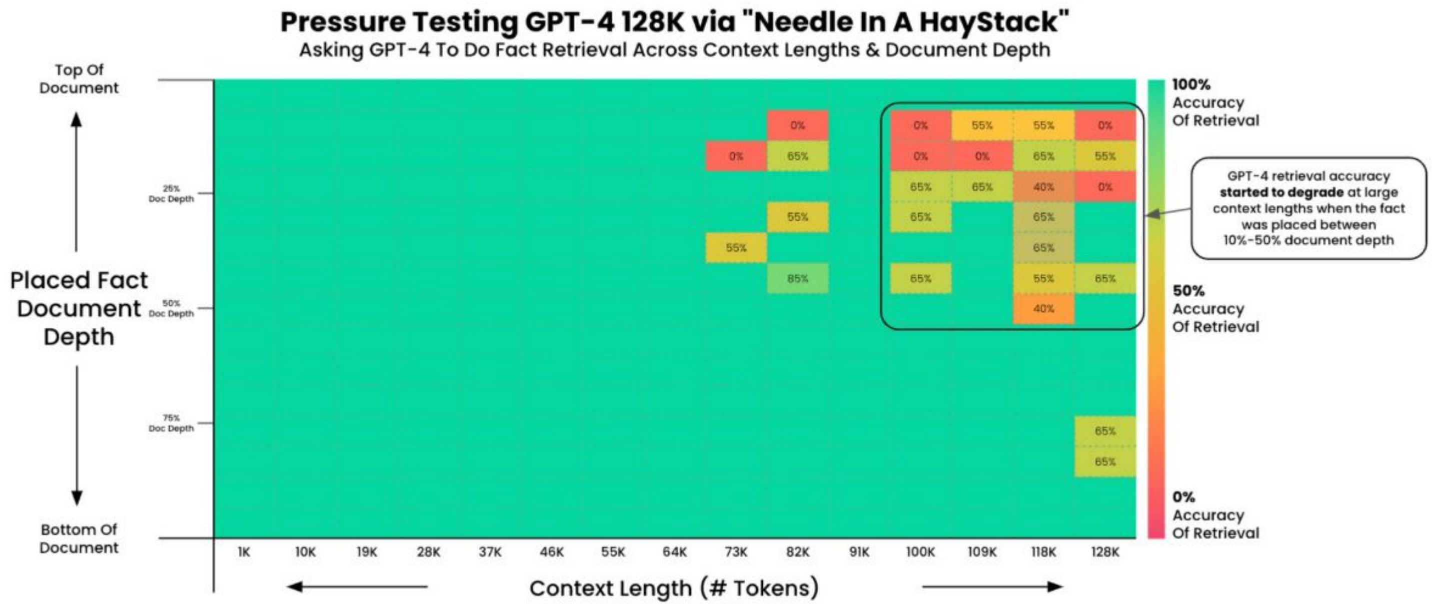
Motivation

- **Limited** context size



Motivation

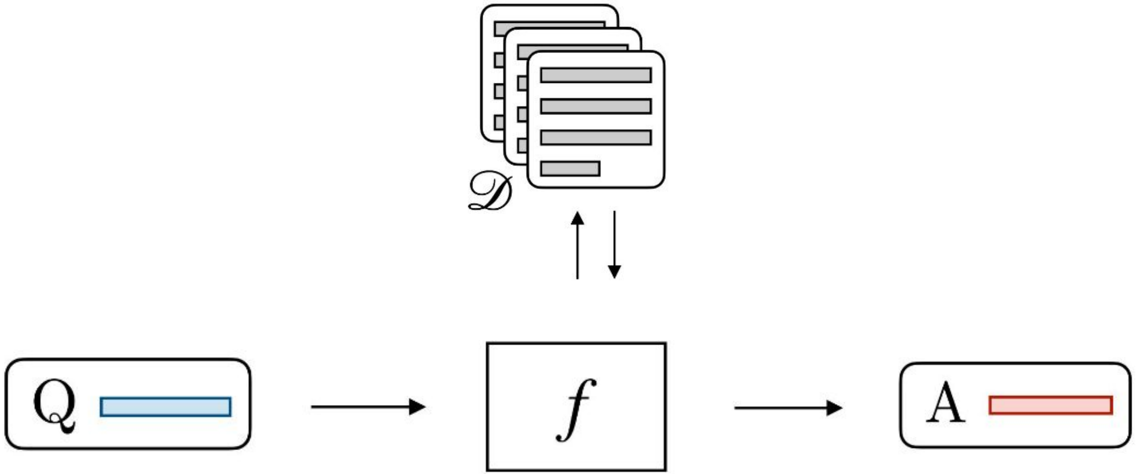
- LLM gets **distracted** by **useless** information



RAG

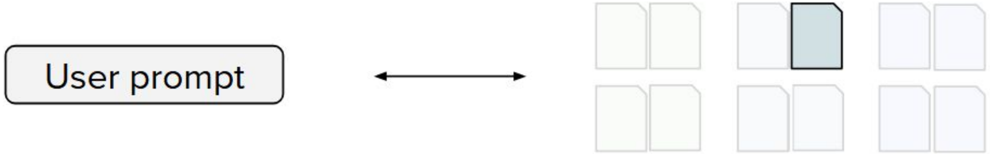
RAG = **R**etrieval-**A**ugmented **G**eneration

Idea. Augment prompt with **relevant** pieces of information.

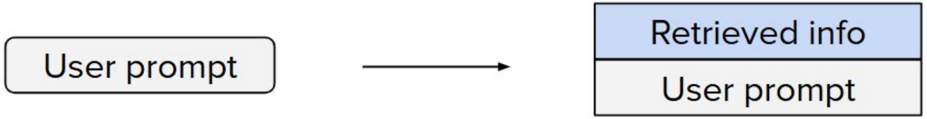


Retrieve, Generate, Augment

1. **Retrieve** relevant document via similarity operation across the knowledge base

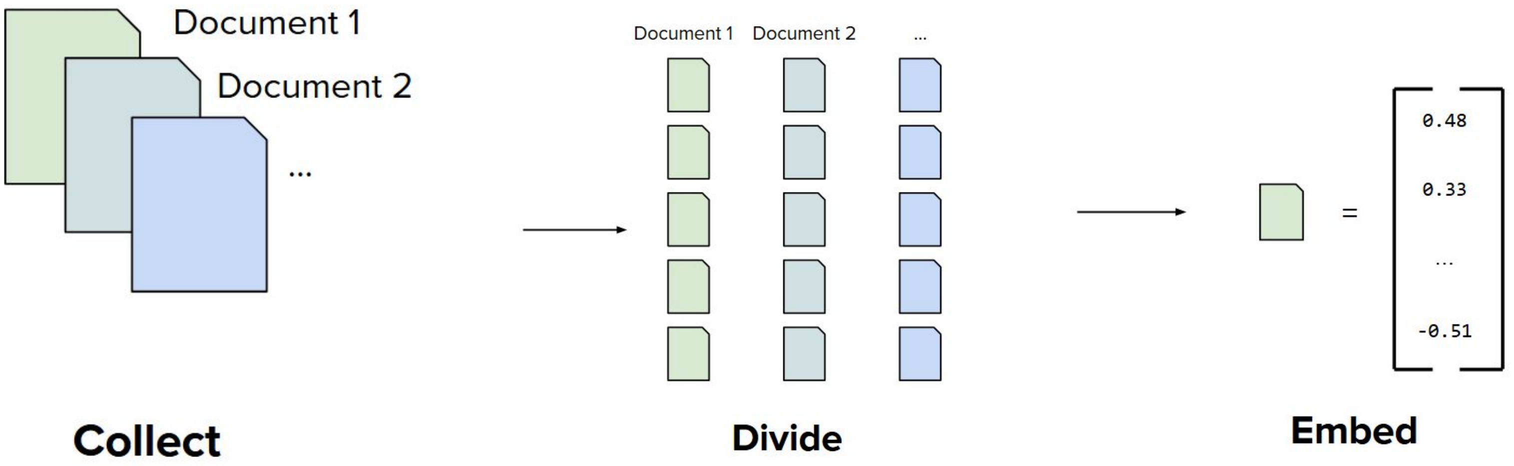


2. **Augment** prompt with retrieved information



3. **Generate** response

Create KB

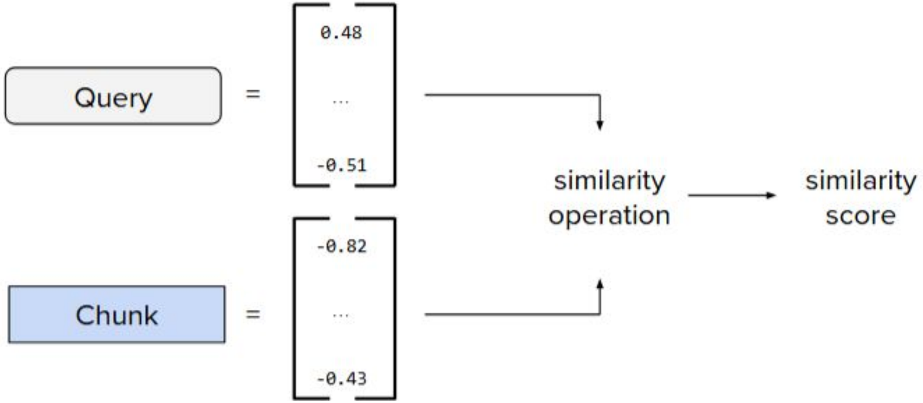


Hyperparameters. Embedding size, chunk size, overlap between chunks

Candidate Retrieval

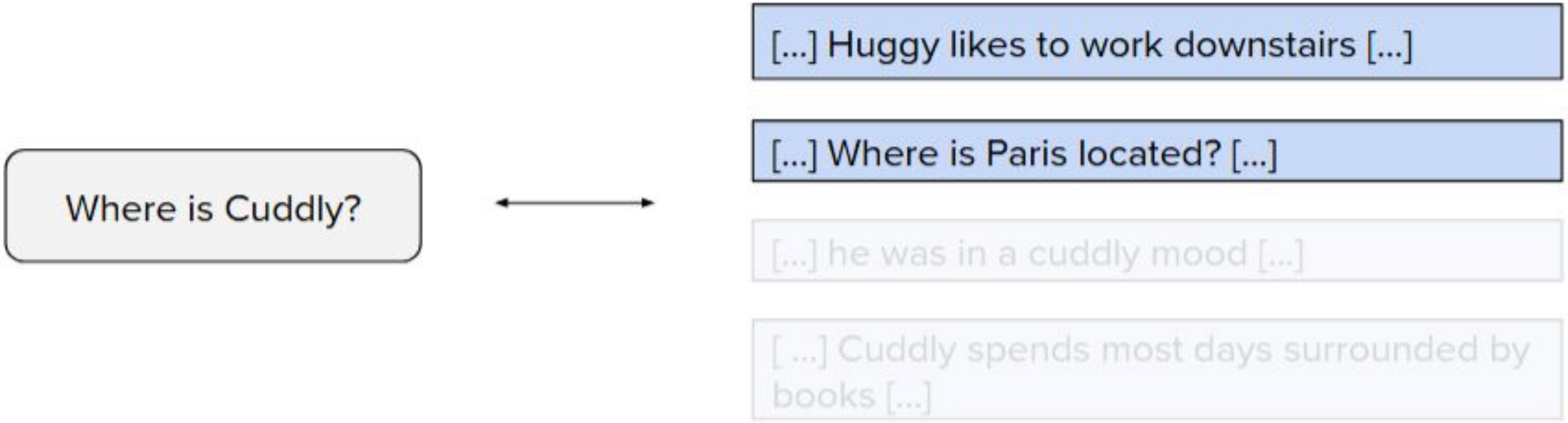
Objective. Select potentially-relevant candidates via search across knowledge base

Method 1. Semantic search using **embeddings-based similarity**



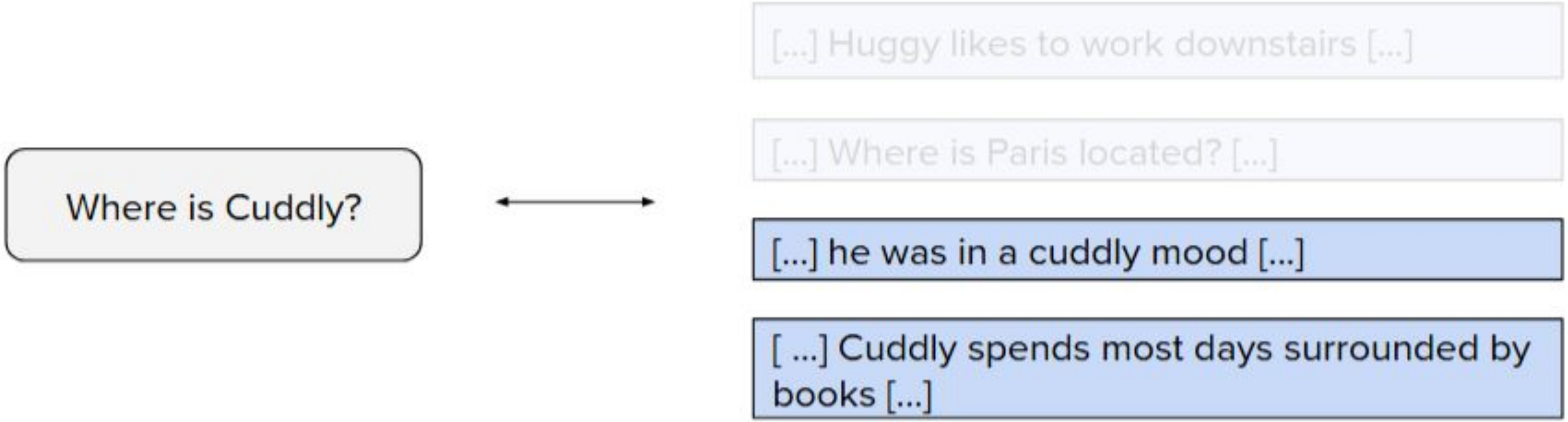
Candidate Retrieval

Method 1. Semantic search using **embeddings-based similarity**



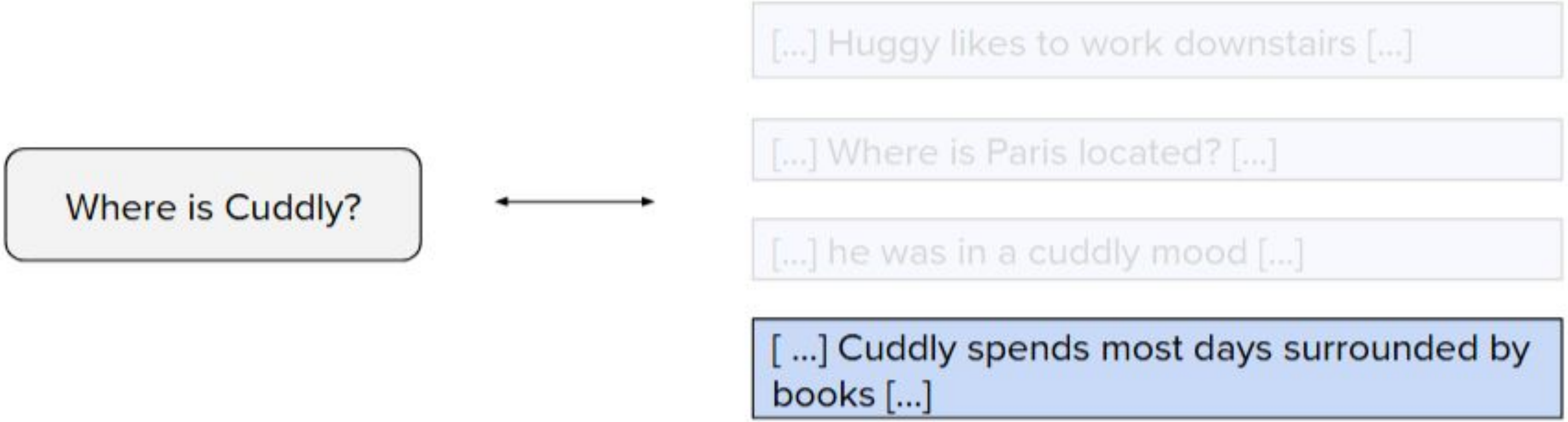
Candidate Retrieval

Method 2. Search based on keywords matching via **BM25**



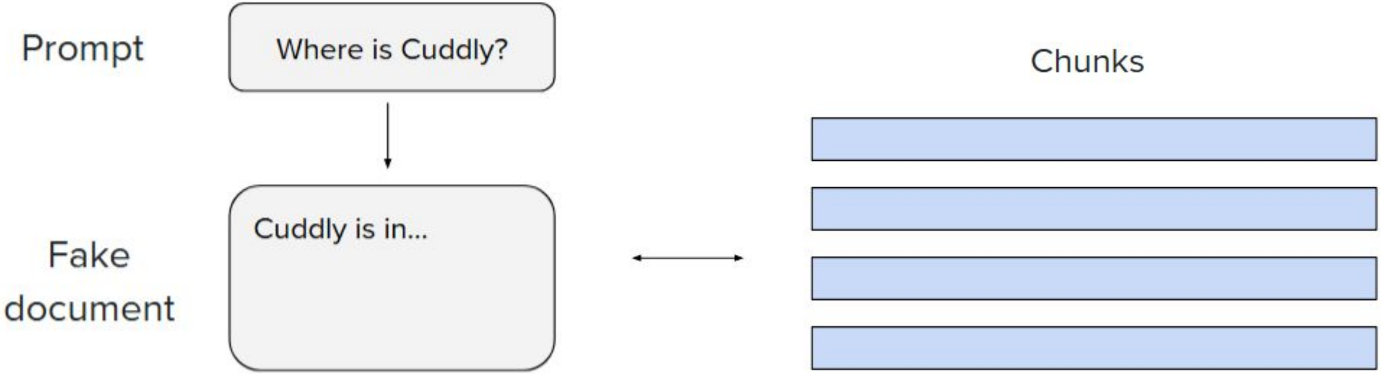
Candidate Retrieval

Method 3. Search based on **hybrid combination** of semantics and BM25



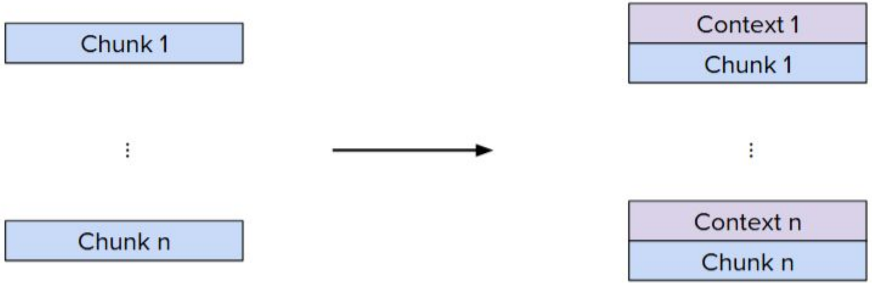
Some More Methods

- Mitigate **discrepancy** in nature of embeddings



Some More Methods

- **Contextualize** document chunks



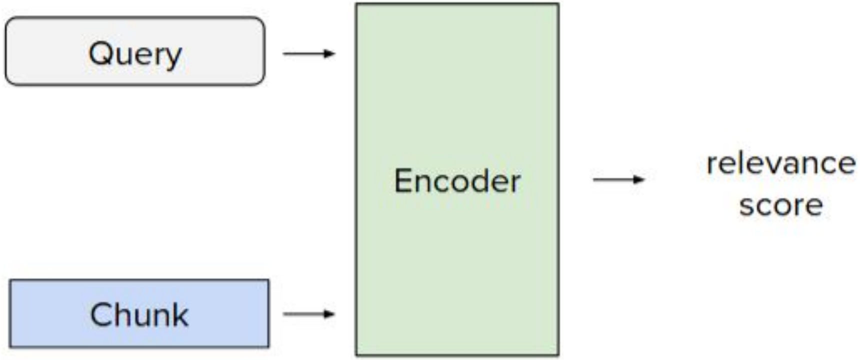

```
<document>
{WHOLE_DOCUMENT}
</document>
Here is the chunk we want to situate within the whole document:
{CHUNK_CONTENT}


Please give a short succinct context to situate this chunk within the overall
document for the purposes of improving search retrieval of the chunk. Answer
only with the succinct context and nothing else.
```

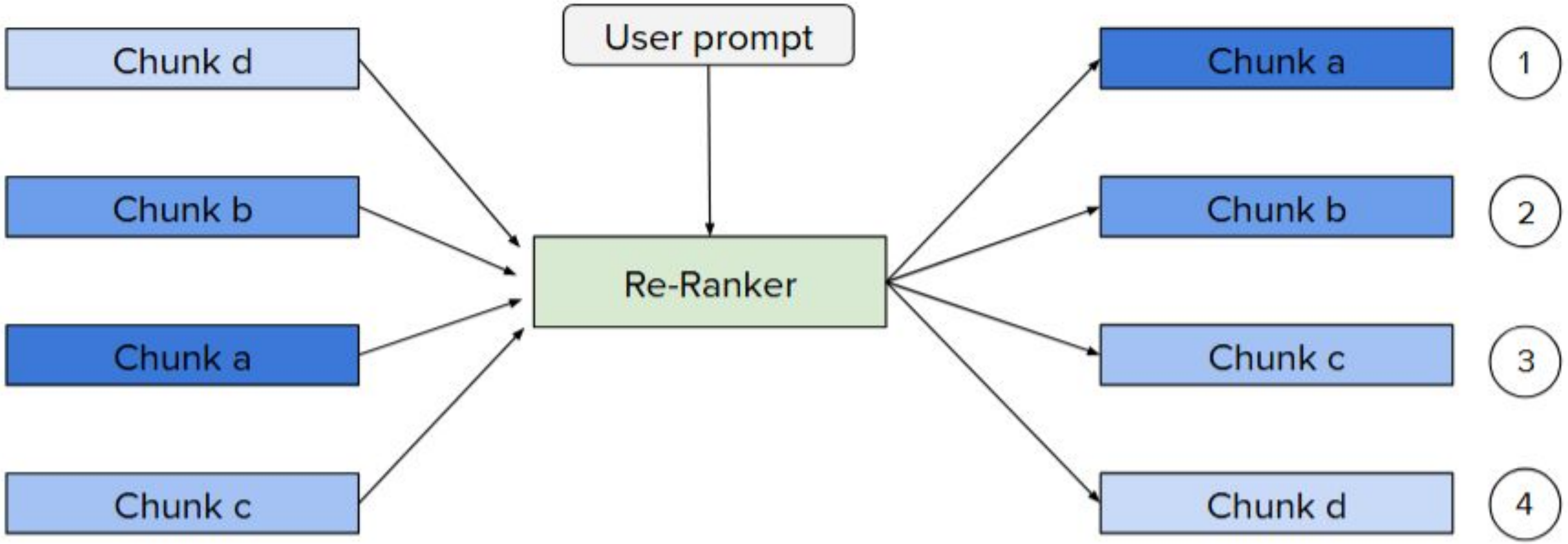
Ranking

Objective. Provide final relevance score using more sophisticated (re-)ranker

"cross-encoder"



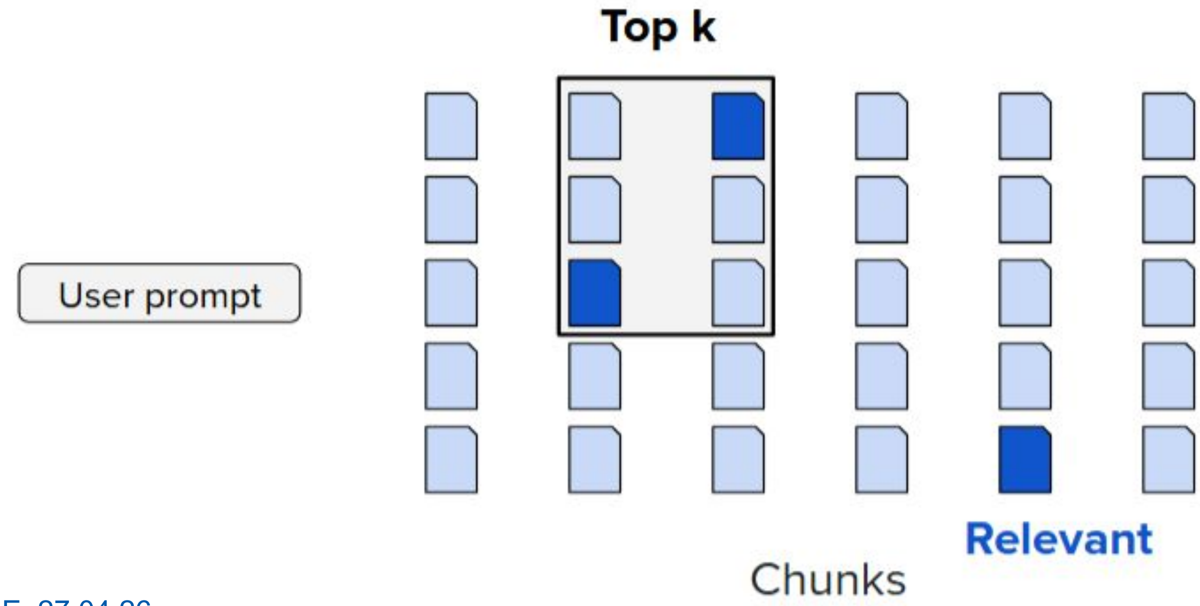
Ranking



Chunk d
Chunk b
Chunk a
Chunk c
User prompt
Re-Ranker
Chunk a
1
Chunk b
2
Chunk c
3
Chunk d
4

Ranking

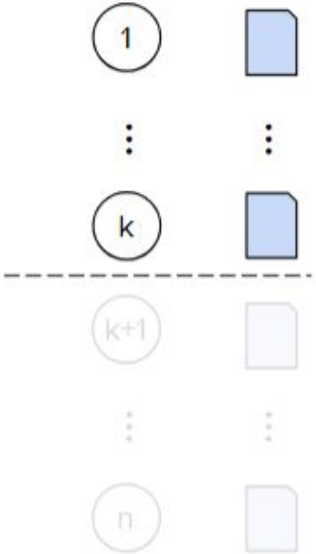
Setup. Evaluate if **retrieved chunks** are **relevant**



Ranking

Setup. Evaluate if **retrieved chunks** are **relevant**

Ranking



- **N**ormalized **D**iscounted **C**umulative **G**ain at **k** (**NDCG@k**)

$$\boxed{\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)}} \quad \text{with } \mathrm{rel}_i \in \{0, 1\}$$

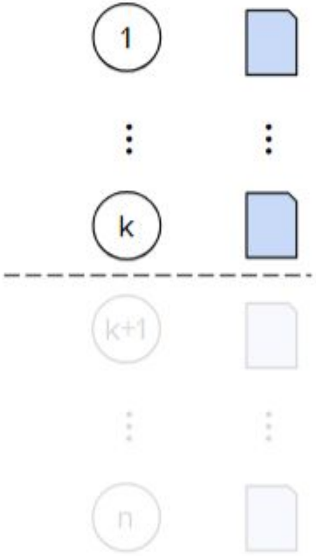
$$\mathrm{NDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k}$$

$\mathrm{DCG@}k$ if ranking was perfect

Ranking

Setup. Evaluate if **retrieved chunks** are **relevant**

Ranking



- **N**ormalized **D**iscounted **C**umulative **G**ain at **k** (**NDCG@k**)
- **R**eciprocal **R**ank at **k** (**RR@k**)

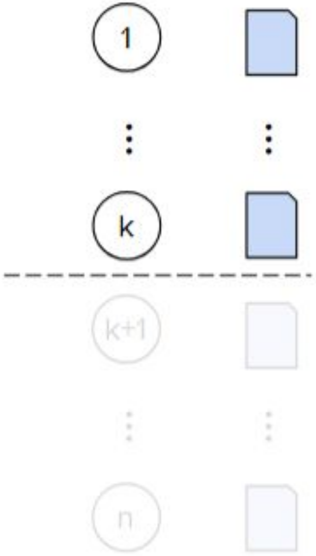
$$\mathrm{RR} = \frac{1}{\mathrm{rank}}$$

rank of the first relevant chunk

Ranking

Setup. Evaluate if **retrieved chunks** are **relevant**

Ranking



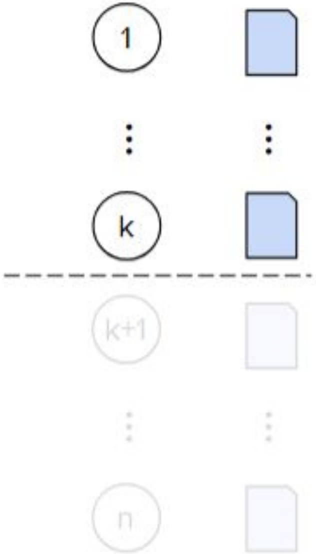
- **N**ormalized **D**iscounted **C**umulative **G**ain at **k** (**NDCG@k**)
- **R**eciprocal **R**ank at **k** (**RR@k**)
- **Recall at k**

$$\boxed{\text{Recall@}k = \frac{|\text{relevant in top } k|}{|\text{relevant}|}}$$

Ranking

Setup. Evaluate if **retrieved chunks** are **relevant**

Ranking



- **N**ormalized **D**iscounted **C**umulative **G**ain at **k** (**NDCG@k**)
- **R**eciprocal **R**ank at **k** (**RR@k**)
- **Recall at k**
- **Precision at k**

$$\text{Precision@}k = \frac{|\text{relevant in top } k|}{k}$$

Tools

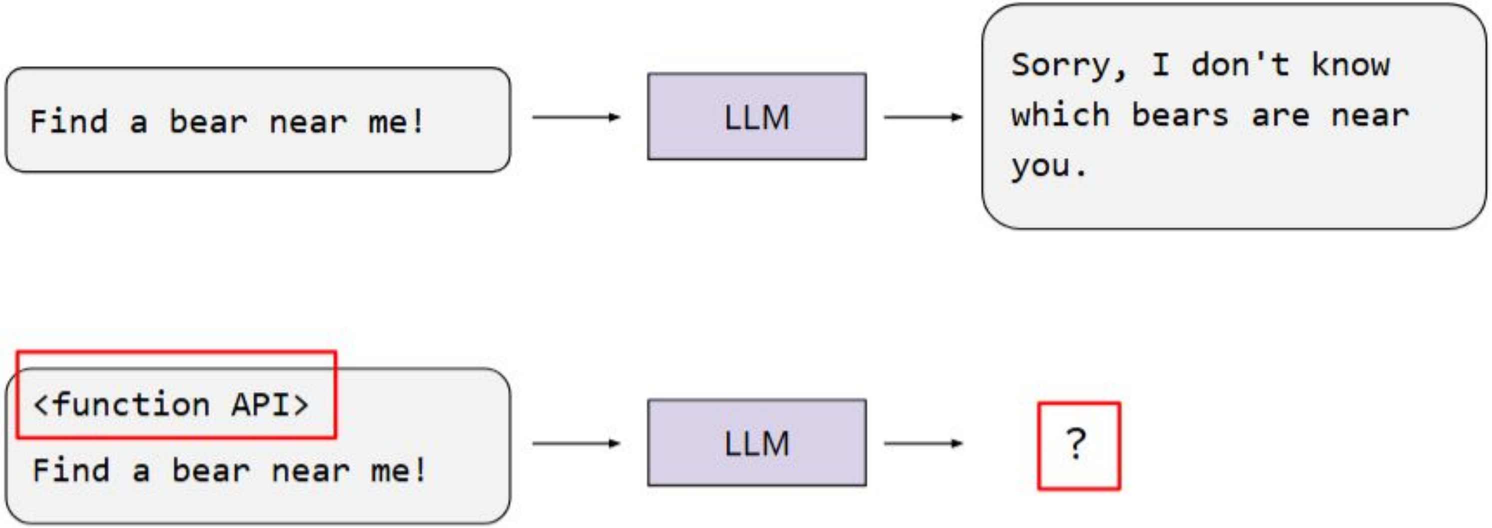
Tools

- Web Search
- Math commuting
- Image Generation
- API Calls

Example

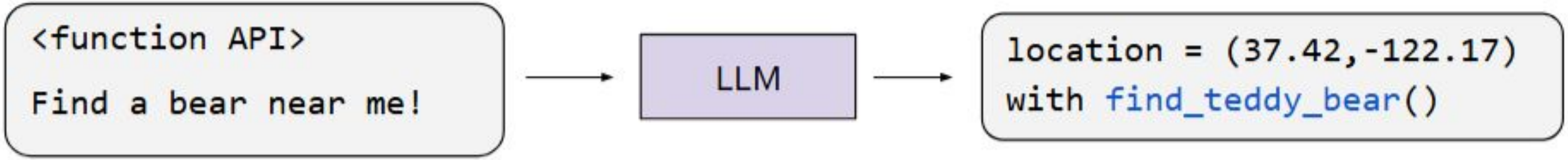
LLM as we've known it so far

LLM with tools



Example

1. Let **LLM** find **argument** for **relevant function** call

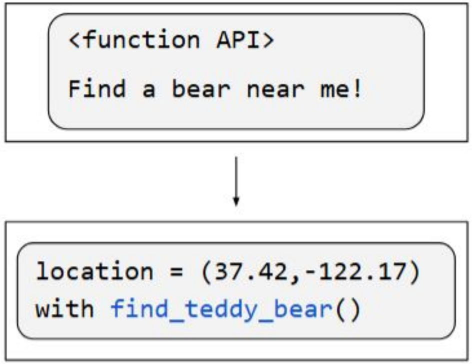


2. Make **function call**

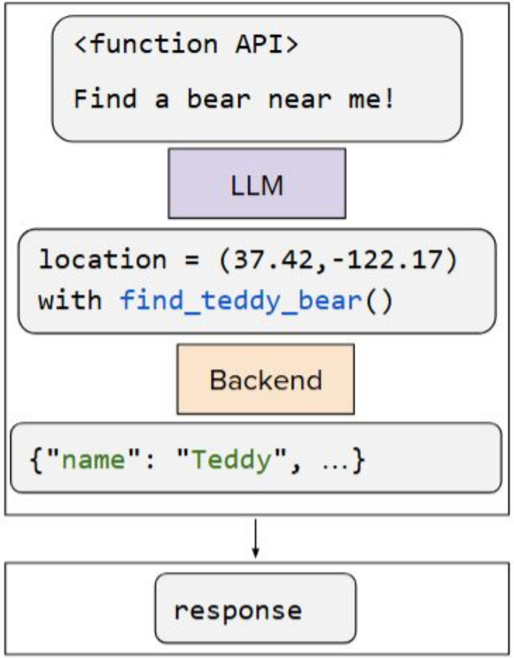
3. Let **LLM deduce** conclusion **based on results**

How To Use Tools?

- Training
- Prompting



Tool prediction



Response generation

```
<function API> + <detailed explanation on how to use it>

Find a bear near me!
```

Summary

PROS

- LLMs just became way more useful!
- They can also interact with the real world
- Overcomes "knowledge cutoff" limitation

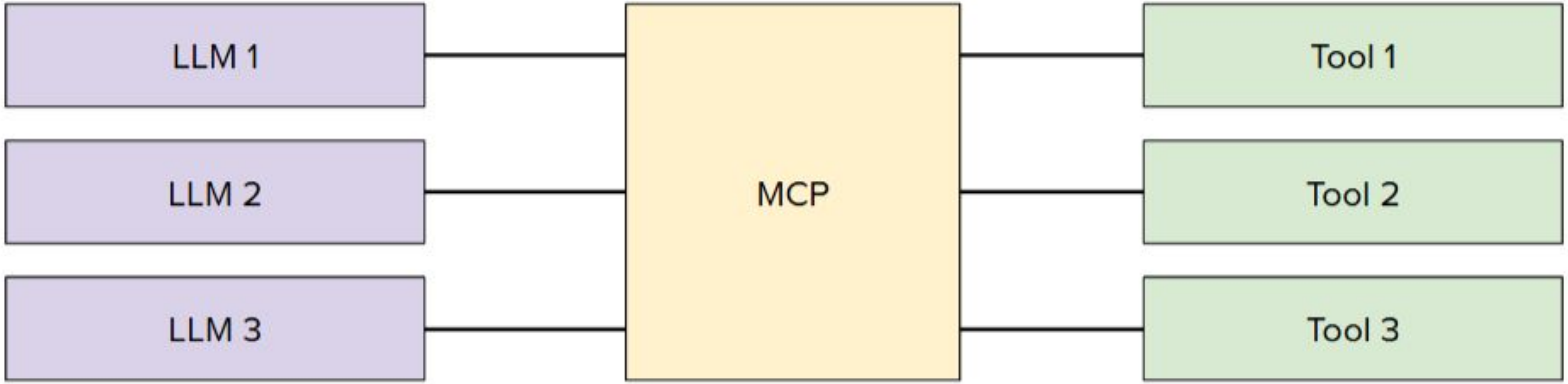
CONS

- More tools = decrease performance
- Finite context length: not scalable
- Many tools to define. Lots of work

MCP

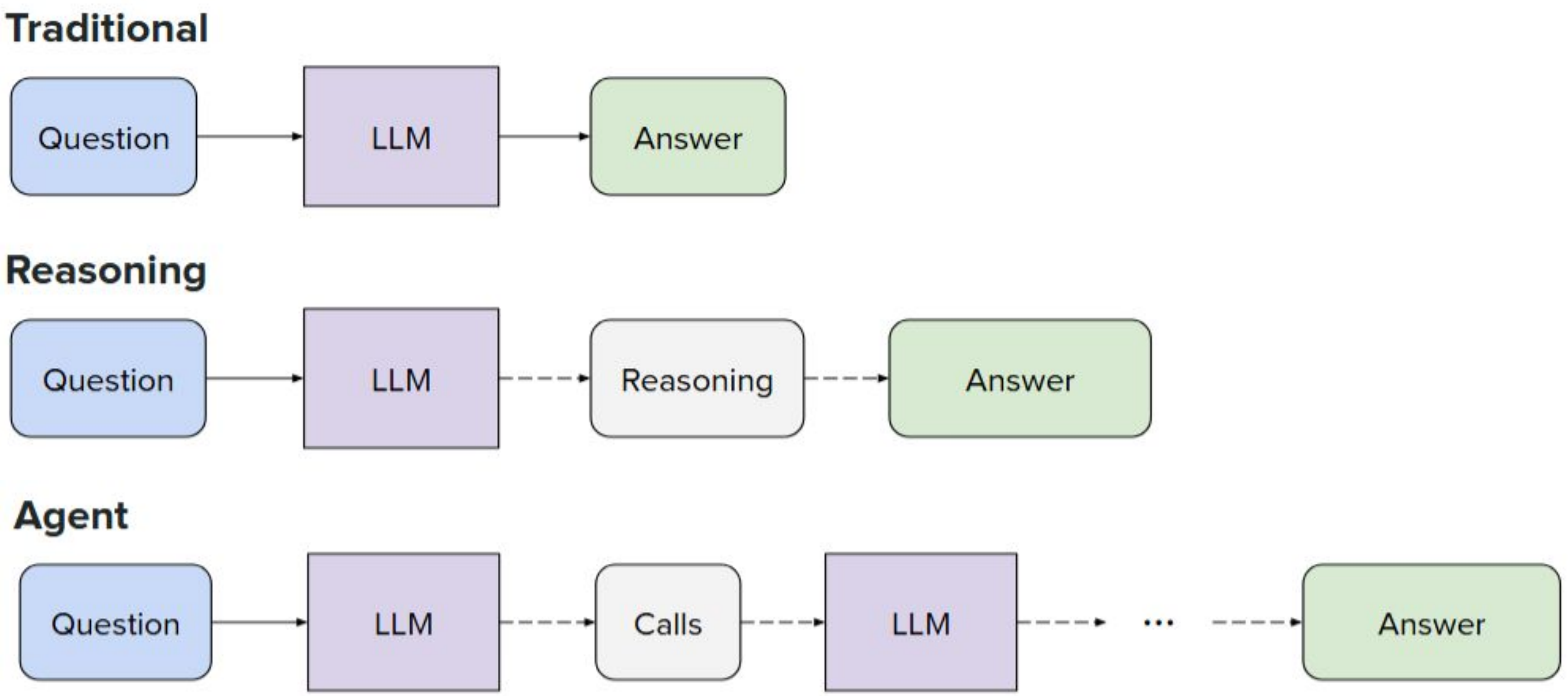
MCP = **M**odel **C**ontext **P**rotocol

Idea. Connect tools/data to LLMs in a standard way

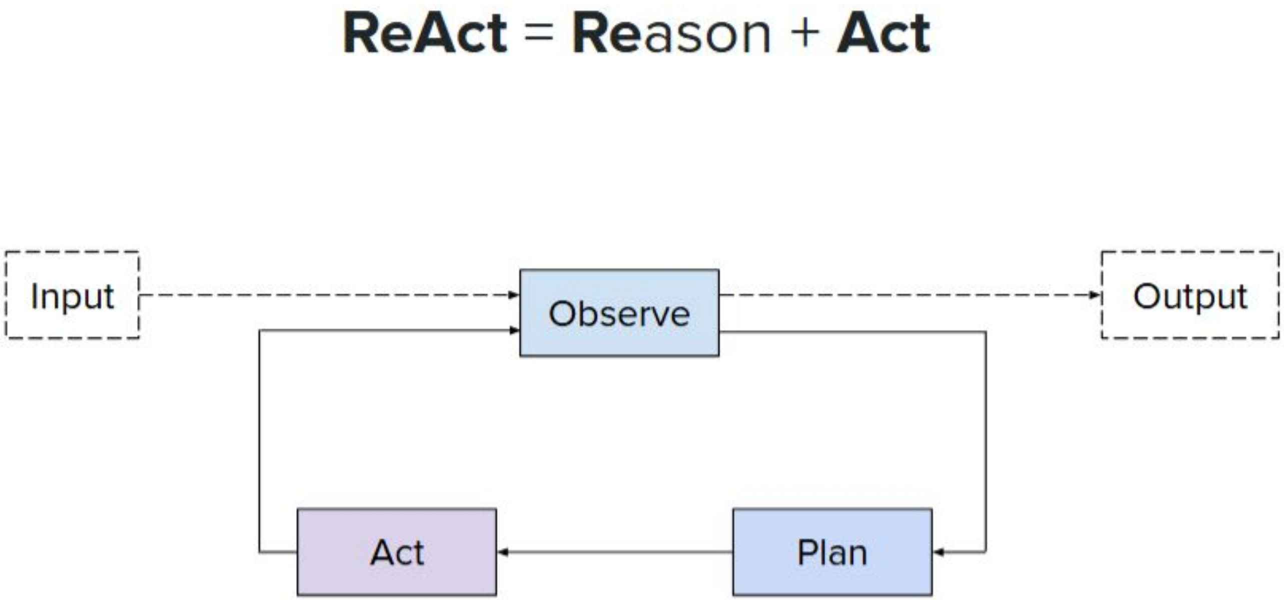


Agents

Idea


Traditional
Question
LLM
Answer
Reasoning
Question
LLM
Reasoning
Answer
Agent
Question
LLM
Calls
LLM
...
Answer

ReAct



ReAct

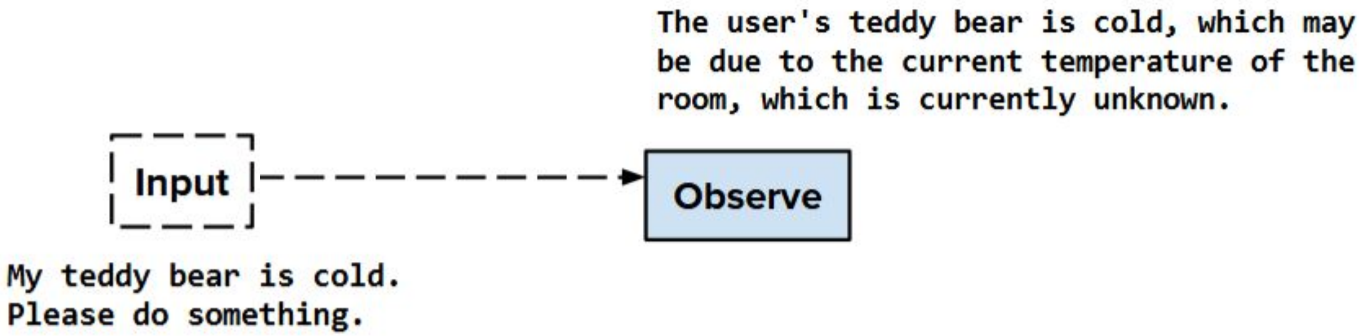
Input

```
My teddy bear is cold.
Please do something.
```

Examples:

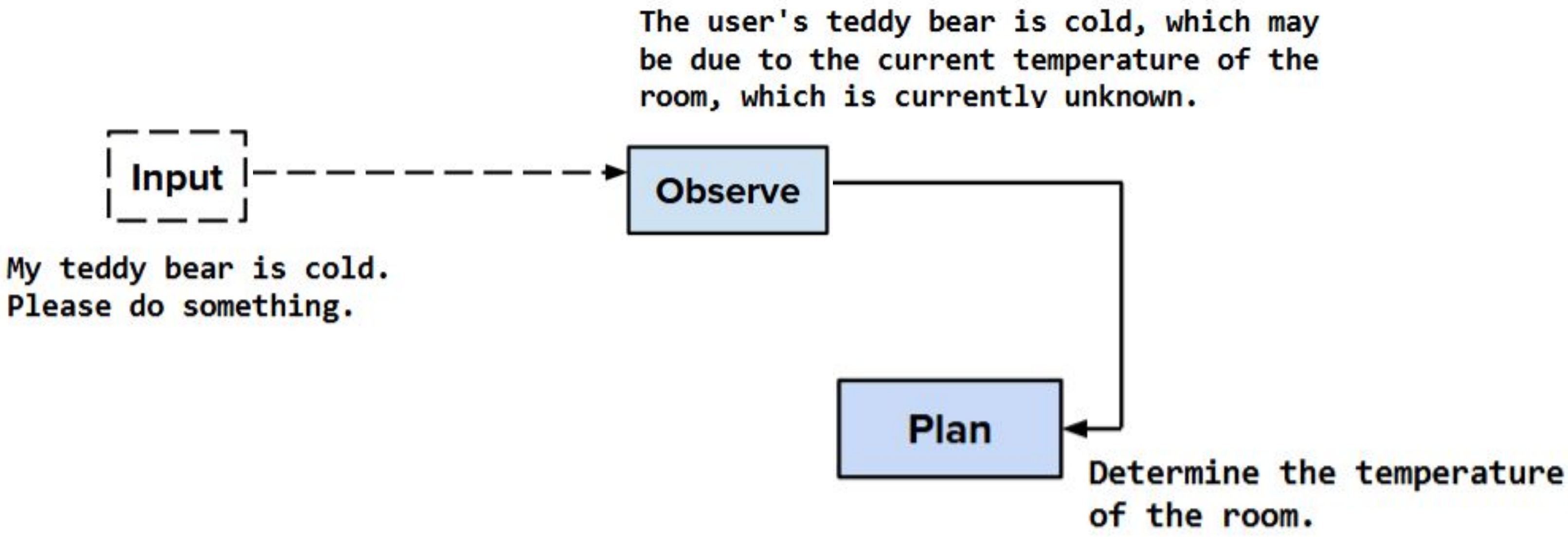
- Manually entered (e.g. user question)
- External event (e.g. metric going beyond a threshold)

ReAct



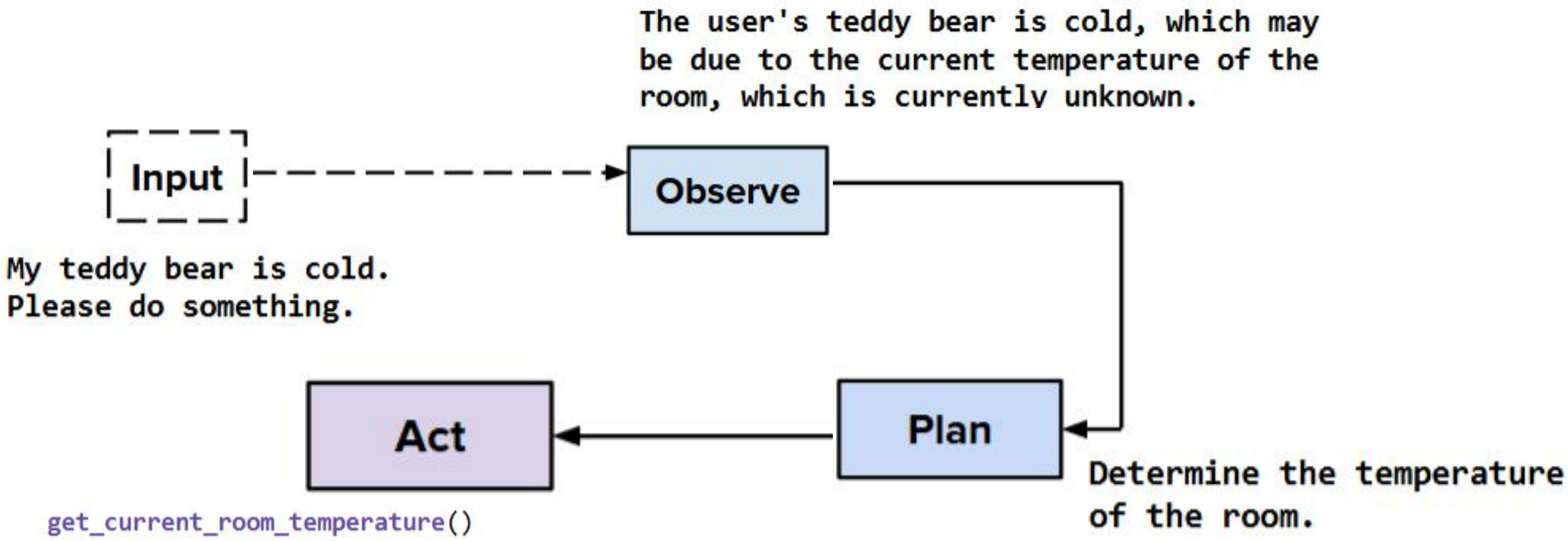
- **Synthesize** previous actions + explicitly **state** what is **currently known** including own knowledge
- **Reasoning**-heavy step to figure out **what is needed**

ReAct



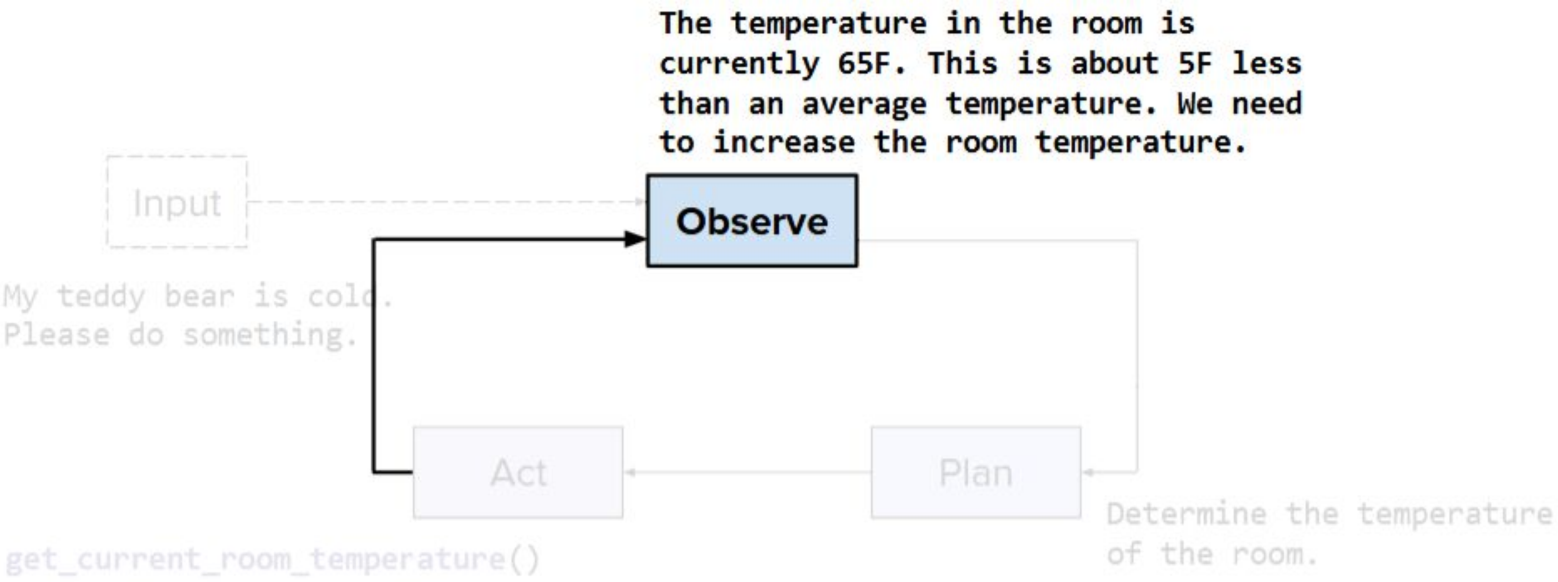
Detail **what tasks** need to be accomplished and **what tools** to call

ReAct

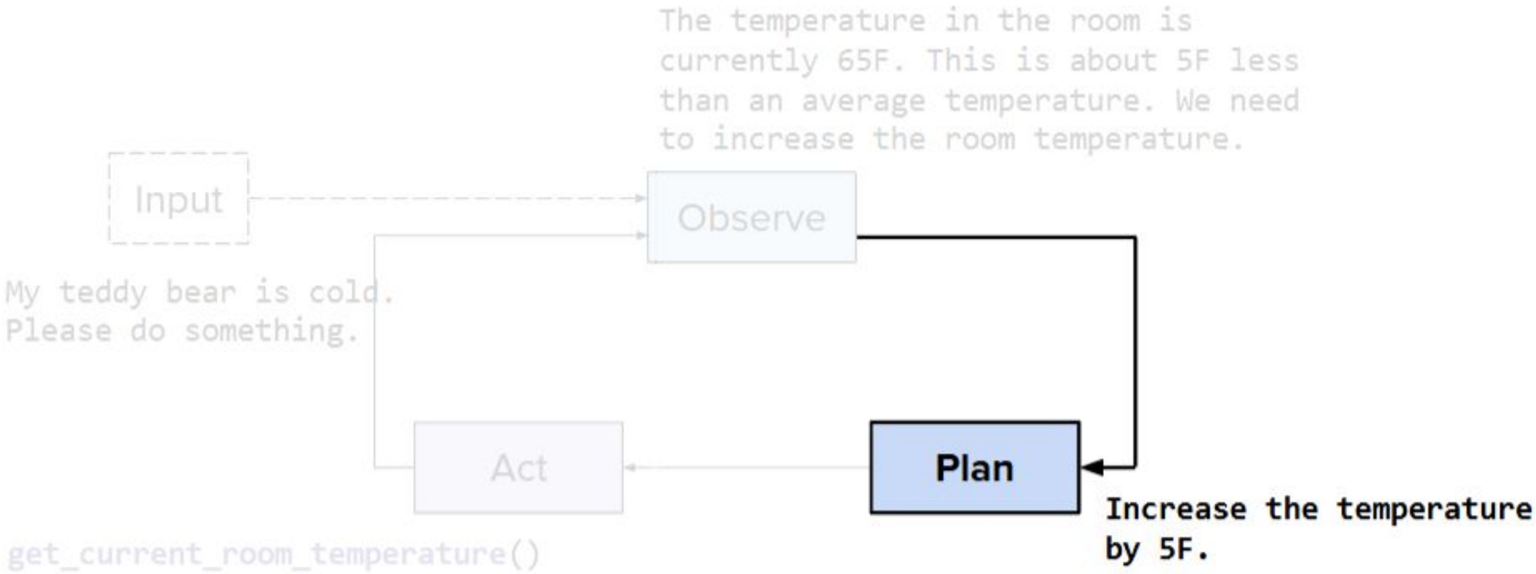


- Perform an action via an **API**
- **Look** for info in a **database** of documents

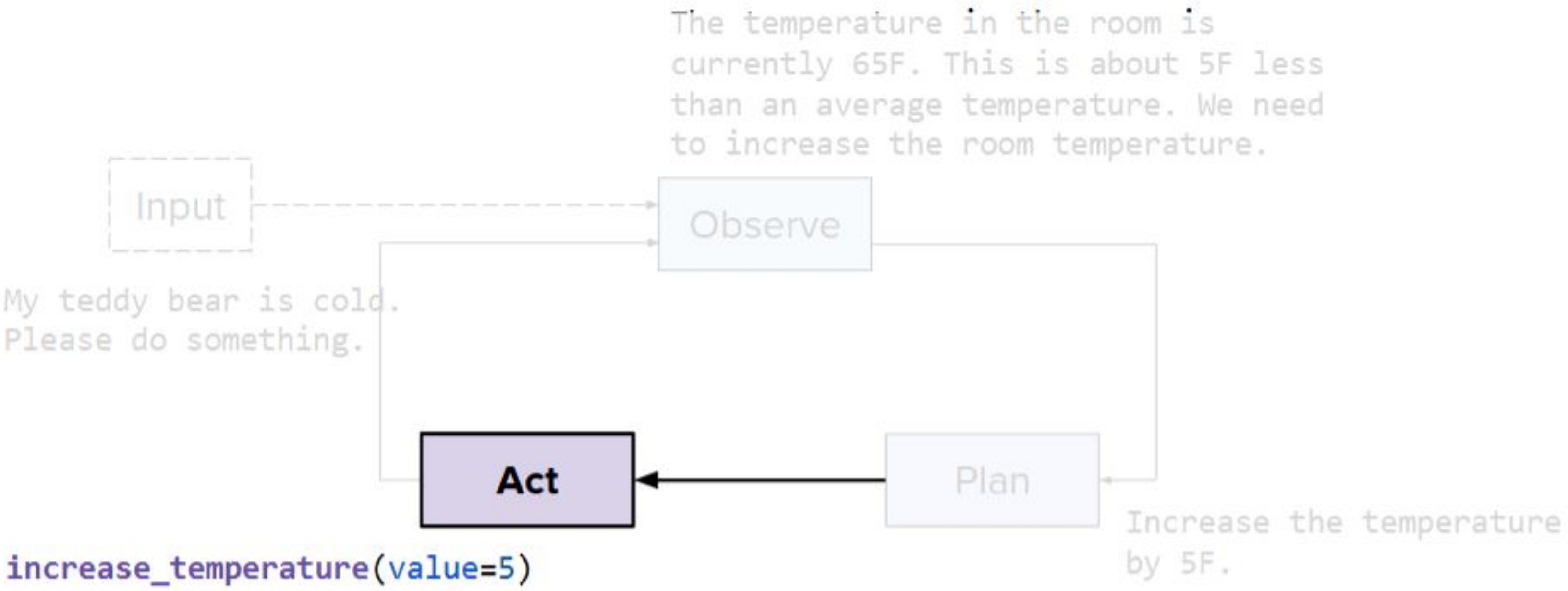
ReAct


The temperature in the room is currently 65F. This is about 5F less than an average temperature. We need to increase the room temperature.
Input
Observe
My teddy bear is cold.
Please do something.
Act
Plan
Determine the temperature of the room.
get_current_room_temperature()

ReAct

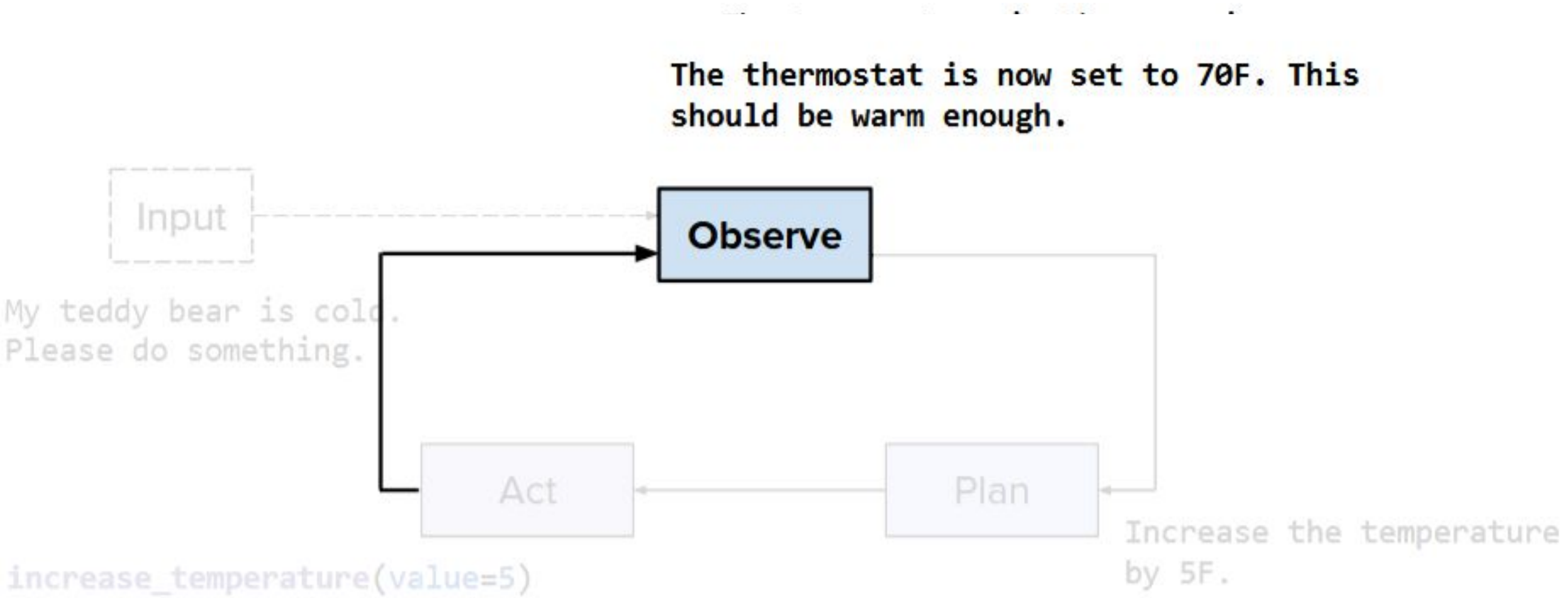

The temperature in the room is currently 65F. This is about 5F less than an average temperature. We need to increase the room temperature.
Input
Observe
My teddy bear is cold. Please do something.
Act
Plan
Increase the temperature by 5F.
get_current_room_temperature()

ReAct

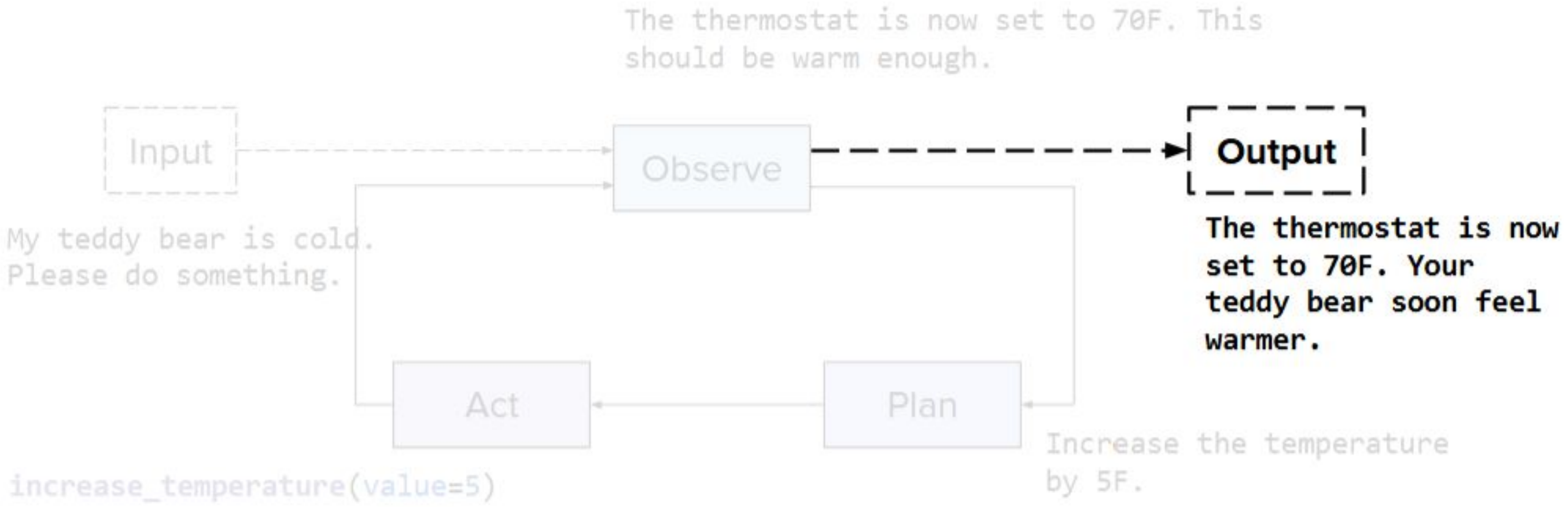


The temperature in the room is currently 65F. This is about 5F less than an average temperature. We need to increase the room temperature.
Input
Observe
My teddy bear is cold. Please do something.
Act
Plan
increase_temperature(value=5)
Increase the temperature by 5F.

ReAct

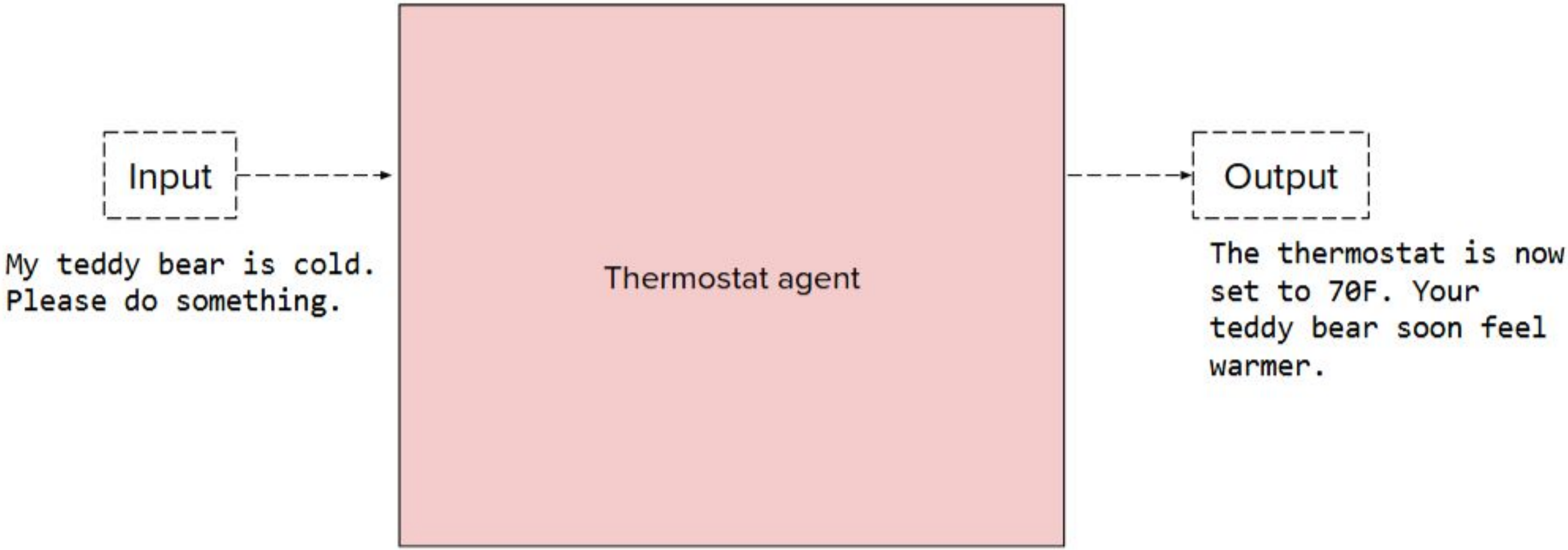

The thermostat is now set to 70F. This
should be warm enough.
Input
Observe
My teddy bear is cold.
Please do something.
Act
Plan
increase_temperature(value=5)
Increase the temperature
by 5F.

ReAct

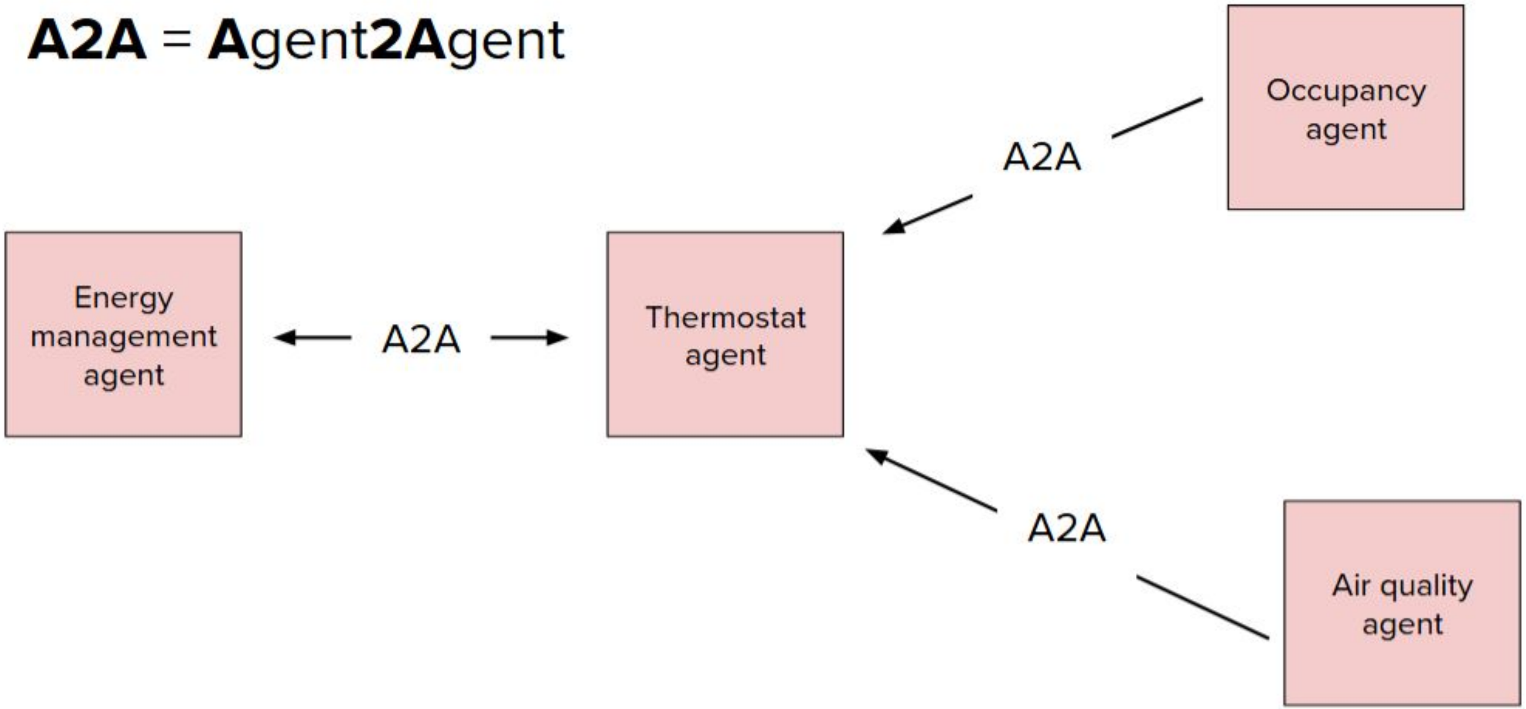


The thermostat is now set to 70F. This should be warm enough.
Input
Observe
Output
My teddy bear is cold. Please do something.
The thermostat is now set to 70F. Your teddy bear soon feel warmer.
Act
Plan
increase_temperature(value=5)
Increase the temperature by 5F.

Agentic View



Agentic View



Safety

Risks.

- Potential for harm in the real world
- Example: data exfiltration

Remediations.

- Training steps
- Inference safeguards
- Benchmarks, e.g. Agent-SafetyBench

...very important topic!

Summary

- **Hallucination** is a (big) problem
- **Reasoning** abilities are a bottleneck
  - Finetuning helps, but hard
  - New capabilities are very welcome
- **Evaluation** is challenging
- Good to start simple, then iterate and progressively scale up
- Good to start with capable models, optimize on size later
- Transparency / observability helps with user trust and debuggability